Software Architecture and Design Specification

Online Bookstore System

Project: Online Bookstore System

Version: 1.0

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Revision History

Version	Date	Author	Change Summary
1	12-09-2026	Team 4	Initial architecture draft derived from Online Bookstore SRS v1.0
2	14-09-2026	Team 4	First complete draft: architecture, design, API, security, traceability

Approvals

Role	Name	Signature/Date
Course Coordinator	Prof Pavithra P	

1. Introduction

1.1 Purpose

This document specifies the software architecture and design of the Online Bookstore System, derived from the Online Bookstore System Software Requirements Specification (SRS), v1.0, dated 09-09-2026.

1.2 Scope

Covers the architecture and design of customer-facing services (registration, login, catalogue browsing and search, cart, checkout, online payment, order history and status) and administrator-facing services (book, category, inventory and order administration, and sales/inventory reporting). Internal operation of the external payment and delivery services is outside the scope of this system, consistent with SRS §1.2 and §2.5.

1.3 Audience

Developers, QA engineers, security auditors, the course guide/evaluators, and future maintenance teams.

1.4 Definitions

SRS, FR, NFR, UI, API, ISBN, RBAC, RTM, TLS, ADR (Architecture Decision Record), STRIDE, ORM (Object-Relational Mapping).

2. Document Overview

2.1 How to use this document

This document describes the architecture and detailed design of the Online Bookstore System. It should be read together with the Online Bookstore System SRS v1.0. The architecture, component descriptions, design flows, APIs and security decisions provide implementation guidance for the requirements defined in the SRS.

2.2 Related Documents

SRS, STP, RTM.

3. Architecture

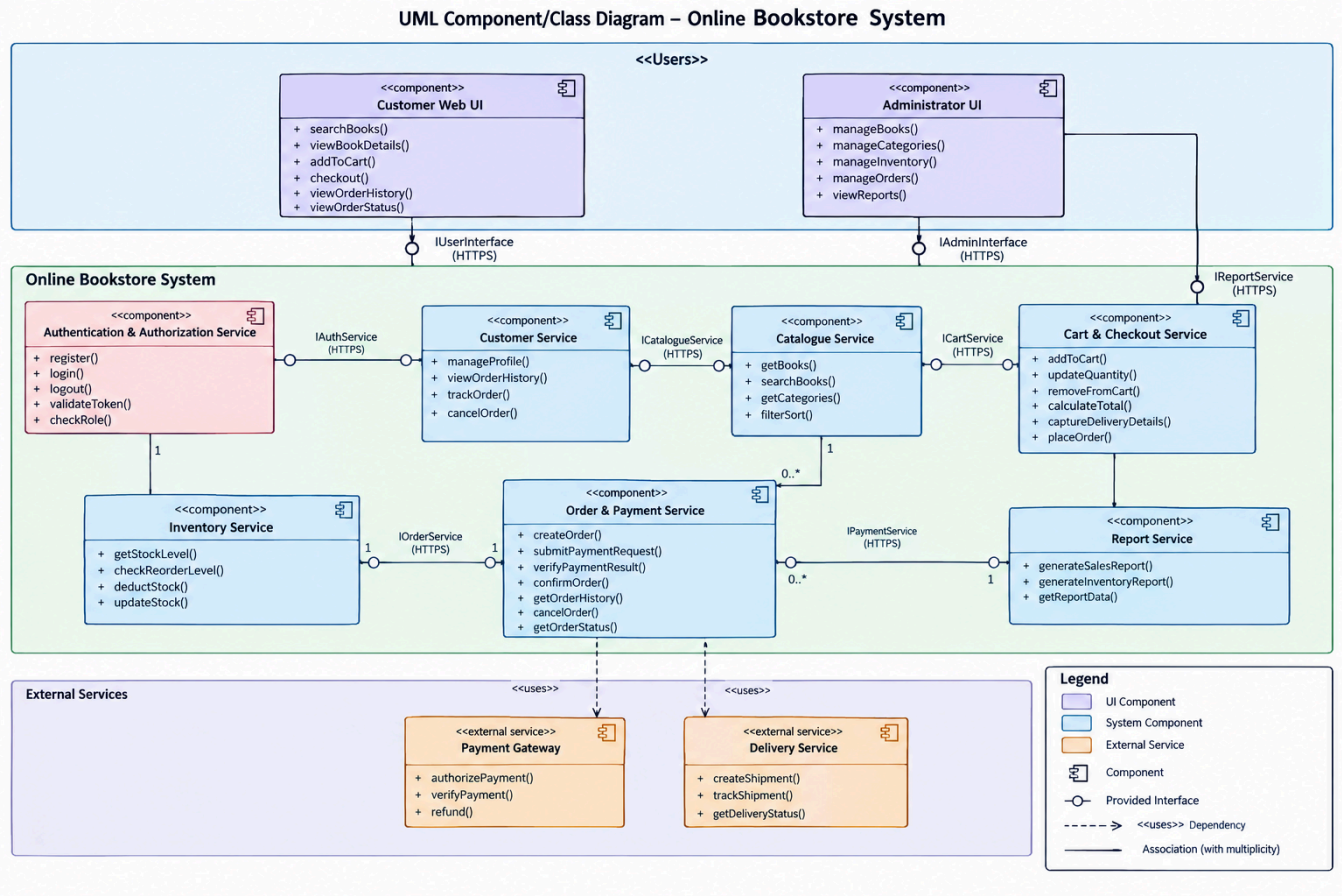
3.1 Goals

- Security: authenticated, role-separated access to customer and administrator functions (OBS-SR-001 to OBS-SR-006).
- Reliability: at least 99.5% monthly availability, excluding planned maintenance (OBS-NF-002).
- Performance: 90th-percentile page response time ≤ 3 seconds under normal load (OBS-NF-001).
- Scalability: support at least 10,000 book records and 20 concurrent customer sessions (OBS-NF-004).
- Usability: a first-time customer can search and add a book to the cart without training, across desktop, tablet and mobile (OBS-NF-003, OBS-NF-005).
- Maintainability: requirements, design and test cases remain traceable as the system changes.

3.2 Stakeholders & Concerns

- Customers: usability, security of account/payment data, order accuracy.
- Administrators: ease of maintaining books, inventory and orders; accurate reports.
- Payment Service: a well-formed, verifiable payment request/response contract.
- Course guide / evaluators: traceability, compliance with SRS and academic rubric.
- Developers: modularity and a scope achievable within the project timeline.

3.3 Component (UML) Diagram



3.4 Component Descriptions

Component	Responsibility
Customer Web UI	Browsing, search, book detail, cart, checkout, delivery details, order history and status for customers (OBS-F-004 to OBS-F-013, OBS-F-018, OBS-F-019).
Administrator UI	Book, category, inventory, order-status and report management screens for authorized administrators (OBS-F-020 to OBS-F-026).
Authentication & Authorization Service	Account creation, login/logout, session management and role-based access control (OBS-F-001 to OBS-F-003, OBS-SR-001 to OBS-SR-003).
Catalogue Service	Maintains book and category data; serves browsing, search, filter and sort requests (OBS-F-004 to OBS-F-008, OBS-F-020 to OBS-F-022).
Cart & Checkout Service	Cart contents, stock-aware quantity validation, delivery details capture and order-total calculation (OBS-F-009 to OBS-F-013).

Component	Responsibility
Order & Payment Service	Order creation, payment-request submission, verification of the payment result, order confirmation, history, status and cancellation (OBS-F-014 to OBS-F-019, OBS-SR-006).
Inventory Service	Stock levels, reorder-level checks and stock deduction on confirmed orders (OBS-F-017, OBS-F-023, OBS-F-024).
Reporting Service	Aggregates sales and inventory data into administrator-facing reports for a selected period (OBS-F-026).
External Payment Gateway	Third-party service that authorizes payment and returns a signed payment result; outside system scope but integrated via a secure API.
Database	Persistent storage for accounts, books, categories, inventory, carts, orders, payments and audit records.

3.5 Chosen Architecture Pattern and Rationale

A layered architecture (presentation / application service / data access) is chosen, with each service exposed through a REST API consumed by the Customer Web UI and Administrator UI. This keeps the system modular enough to satisfy the traceability and maintainability goals while remaining achievable by a 3-person student team within the project timeline. A microservices split was considered and rejected: at the scale required by the SRS (10,000 books, 20 concurrent sessions) it adds deployment and operational overhead without a corresponding benefit, and would exceed the resource constraint in SRS.

3.6 Technology Stack & Data Stores

- Frontend: a responsive single-page web client (e.g. React) serving both the Customer and Administrator UIs, meeting the multi-device usability requirement (OBS-NF-005).
- Backend: REST APIs (e.g. Spring Boot or an equivalent framework) implementing the services.
- Database: a relational database (e.g. MySQL/PostgreSQL) for accounts, catalogue, inventory, cart, orders, payments and audit records.
- Security: TLS for all client-server traffic; salted one-way password hashing; RBAC enforced server-side.
- External integration: a REST/HTTPS client to the external payment gateway.

3.7 Risks & Mitigations

Risk	Mitigation
External payment service is slow, unreachable, or returns an ambiguous result.	Verify the payment result before marking an order as paid; a failed or missing response leaves the order unpaid rather than confirmed, per OBS-SR-006 and the communications requirement in SRS.
Concurrent purchases of the same book oversell stock.	Stock check and decrement performed as a single transactional operation in the Inventory Service at order confirmation (OBS-F-011, OBS-F-017).
Catalogue and search pages miss the 3-second response target under peak load (up to 20 concurrent sessions, 10,000 book records).	Read-optimized catalogue queries, indexing on title/author/ISBN/category, and caching of frequently browsed catalogue pages (OBS-NF-001, OBS-NF-004).

Risk	Mitigation
Student-team scope and timeline limit what can be built and tested before submission.	Layered architecture with clearly separated services so features can be implemented and demonstrated incrementally, prioritizing High-priority requirements first.

3.8 Traceability to Requirements

Requirement(s)	Description	Component
OBS-F-001 – OBS-F-003OBS-SR-001 – OBS-SR-003	Account creation, login, logout, authentication and RBAC	Authentication & Authorization Service
OBS-F-004 – OBS-F-008OBS-F-020 – OBS-F-022	Catalogue browsing, search, filter, sort, book detail, book/category maintenance	Catalogue Service
OBS-F-009 – OBS-F-013	Cart management, stock-checked quantities, delivery details, order total	Cart & Checkout Service
OBS-F-014 – OBS-F-019OBS-SR-006	Payment request/verification, order confirmation, history, status, cancellation	Order & Payment Service
OBS-F-017, OBS-F-023, OBS-F-024	Stock deduction, stock/reorder-level updates, low-stock identification	Inventory Service
OBS-F-025	Administrator order-status update	Order & Payment Service (admin-authorized)
OBS-F-026	Sales and inventory reports for a selected period	Reporting Service
OBS-NF-001, OBS-NF-004	Response-time and scalability targets	Catalogue Service, Web/App tier, caching
OBS-NF-002	99.5% monthly availability	All services / deployment redundancy
OBS-NF-003	First-time customer can search and add books without training	Customer Web UI
OBS-NF-005	System is usable on desktop, tablet and mobile	Customer Web UI / Administrator UI
OBS-NF-006	Audit trail of administrator changes	Database (audit log) / Admin UI

3.9 Security Architecture

Threat modeling follows STRIDE, aligned to the Security Objectives and Requirements in SRS §5.1:

STRIDE Category	Mitigation	Linked Requirement(s)
Spoofing	Password + session-token authentication; all protected endpoints require a valid session.	OBS-SR-001, OBS-SR-002

STRIDE Category	Mitigation	Linked Requirement(s)
Tampering	Server-side input validation/sanitization on account, search, address and admin forms; parameterized queries.	OBS-SR-005
Repudiation	Audit log of administrator changes to books, inventory and order status, recorded with user, action, record and time.	OBS-NF-006
Information Disclosure	TLS for all account, order and payment-related traffic; passwords stored as salted one-way hashes, never in plaintext.	OBS-SR-002, OBS-SR-004
Denial of Service	Rate limiting on authentication and search endpoints; caching of catalogue reads to protect the 3-second response target under load.	OBS-NF-001, OBS-NF-004
Elevation of Privilege	Role-based access control separating Customer and Administrator functions; server-side role checks on every administrative operation.	OBS-SR-003

4. Design

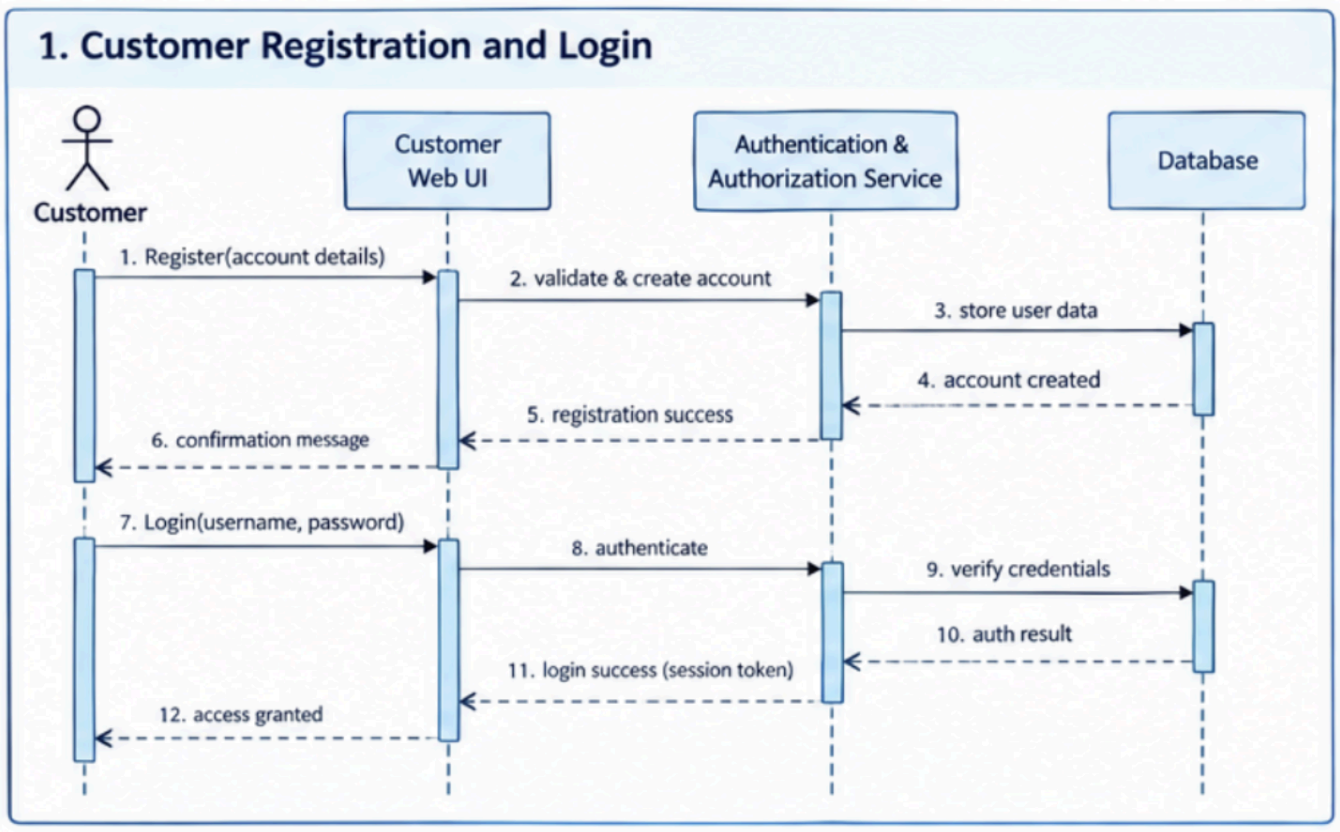
4.1 Design Overview

The Online Bookstore System is designed as layered modules mapping directly to the components in §3.4, so each functional area (authentication, catalogue, cart/checkout, orders/payment, inventory, reporting) can be developed, tested and traced independently while sharing a common database and API conventions.

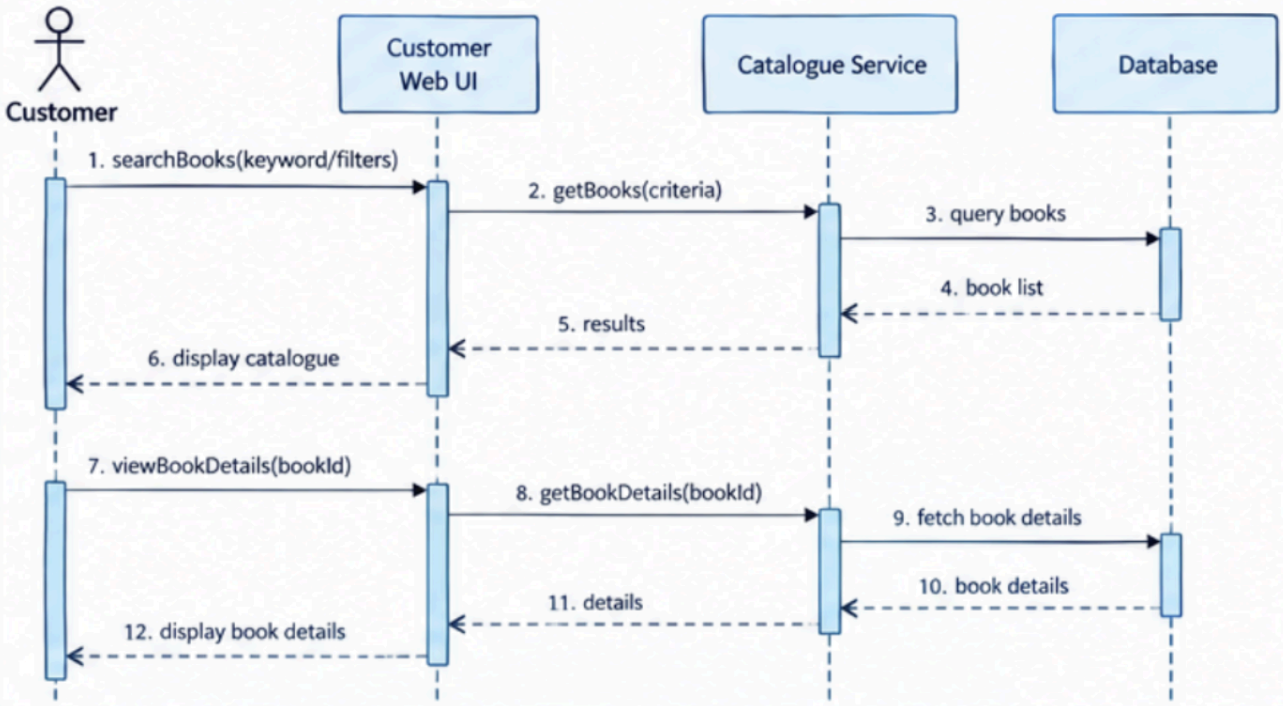
4.2 UML Sequence Diagrams

At least 2 sequence diagrams covering flows specific to this project:

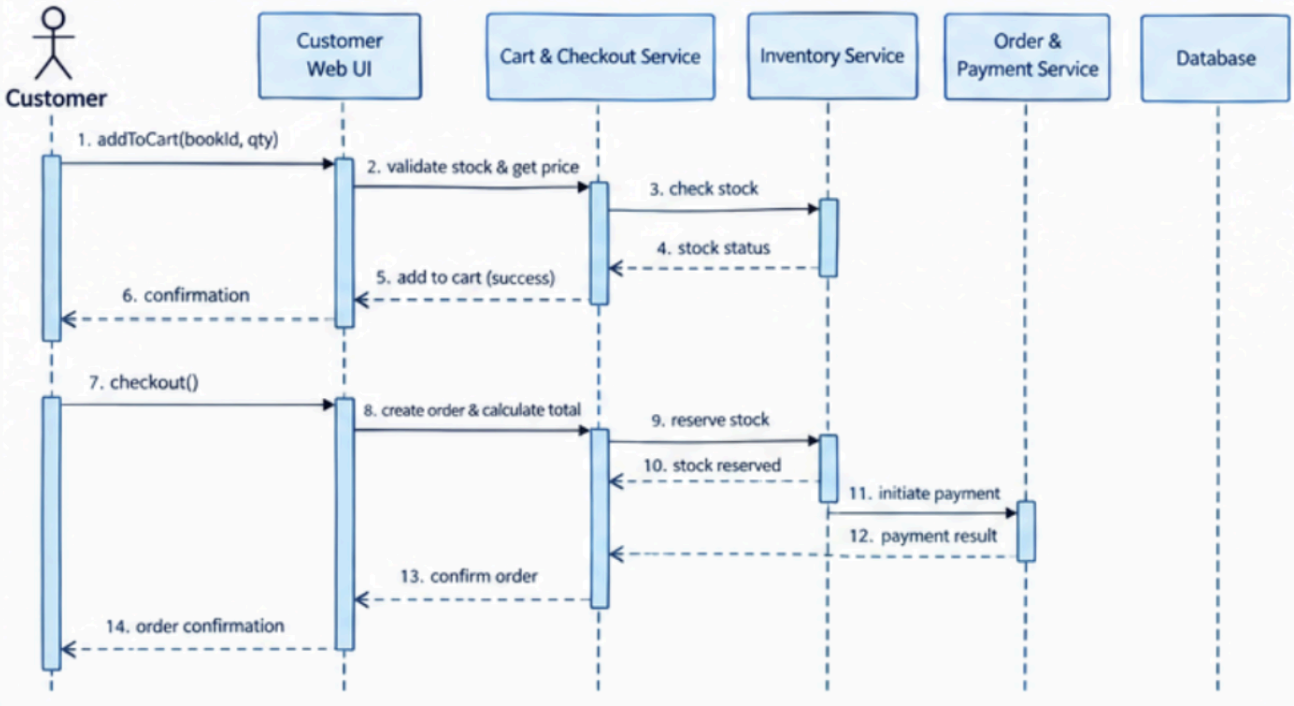
- The system includes two important sequence flows. The checkout and payment flow begins with the Customer interacting with the Cart & Checkout Service, which manages the cart and validates the order details. The request is then passed to the Order & Payment Service, which creates the order and submits the payment request to the External Payment Gateway. After the payment result is verified, the Inventory Service updates and deducts the stock for the confirmed order. This flow covers OBS-F-012 to OBS-F-017 and OBS-SR-006.
- The administrator stock update flow begins when the Administrator uses the Admin UI to update the stock quantity or reorder level of a book. The request is forwarded to the Inventory Service, which updates the corresponding information in the Database and performs a low-stock check. The updated information and any low-stock condition are then reflected back through the Admin UI. This flow covers OBS-F-023 and OBS-F-024.

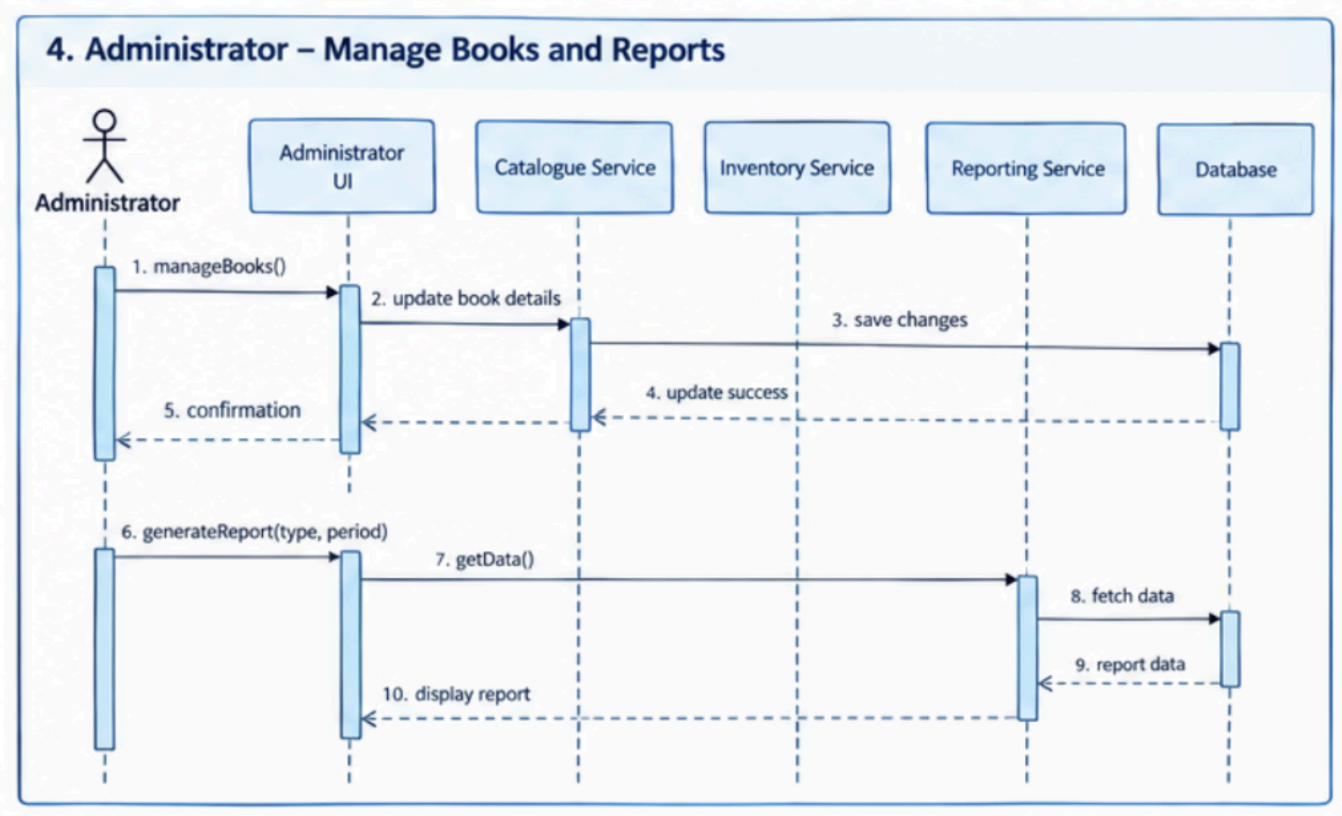


2. Browse and Search Catalogue



3. Cart and Checkout





4.3 API Design

Interface definitions for the Catalogue Service and the Order & Payment Service (at least 2 components):

Catalogue Service — Book Search

Endpoint: /api/books/search?q={query}&category={category}&minPrice={min}&maxPrice={max}&sort={field}

Method: GET

Request: { q, category, minPrice, maxPrice, sort }

Response: { results: [{id, title, author, isbn, price, category, availability}], count }

Errors: 200 OK; 204 No Content when no books match (OBS-F-005, OBS-F-006, OBS-F-007).

Order & Payment Service — Checkout

Endpoint: /api/orders/checkout

Method: POST

Request: { cartId, deliveryAddress, contactDetails }

Response: { orderId, orderNumber, status, total, paymentRequestId }

Errors: 400 Missing/invalid delivery details; 409 Item quantity exceeds available stock (OBS-F-011, OBS-F-012, OBS-F-013).

Order & Payment Service — Payment Result Callback

Endpoint: /api/orders/{orderId}/payment-result

Method: POST

Request: { paymentRequestId, result, signature }

Response: { orderId, status: 'paid' | 'payment_failed' }

Errors: 401 Invalid/unverifiable signature (order is not marked paid); 404 Unknown order (OBS-F-014, OBS-F-015, OBS-SR-006).

4.4 Error Handling, Logging & Monitoring

Standardized error responses (HTTP status + machine-readable code + user-facing message) across all services. No sensitive information (passwords, full card/payment details) is written to logs. Administrator changes to books, inventory and order status are recorded in an audit log with user, action, affected record and timestamp (OBS-NF-006). Monitoring covers catalogue/search response time (OBS-NF-001), monthly availability (OBS-NF-002), transaction/payment failure rate, and low-stock alerts (OBS-F-024).

4.5 UX Design

Customer UI: clear navigation for browsing, search, cart and checkout; validation messages on missing/invalid form input; responsive layout usable without training on desktop, tablet and mobile screen sizes (OBS-NF-003, OBS-NF-005). Administrator UI: consolidated access to books, categories, inventory, orders and reports, restricted to authorized administrators (OBS-SR-003).

4.6 Open Issues & Next Steps

- Notification service for order confirmation and status changes (currently optional per SRS §3.3).
- Caching/CDN strategy to comfortably sustain the 3-second response target as the catalogue grows beyond 10,000 records.
- Detailed data model / ER diagram and index design for the database layer.
- Load-testing plan to validate OBS-NF-004 (10,000 records, 20 concurrent sessions) before acceptance.

5. Appendices

5.1 Glossary

SRS, FR, NFR, UI, API, ISBN, RBAC, RTM, TLS, ADR, STRIDE, ORM.

5.2 References

- Online Bookstore System SRS v1.0 (Team 4).
- IEEE 42010 — Systems and software engineering — Architecture description.
- OWASP Top 10 / OWASP ASVS.
- NIST SP 800-160 — Systems Security Engineering.

5.3 Tools

PlantUML, draw.io, Swagger/OpenAPI, Postman.

5.4 Requirements Traceability Matrix (Excerpt)

Full RTM is maintained in the SRS §8; a representative excerpt is shown here for architecture-review purposes. Status: N = not started, P = partially implemented, A = accepted.

Req ID	Requirement short	Module	Test case(s)	Status
OBS-F-002	Registered customer login with valid credentials	Authentication	TC-Auth-02	N
OBS-F-009	Add available book and quantity to cart	Cart / Checkout	TC-Cart-01	N
OBS-F-015	Record payment successful only after valid response	Payment / Orders	TC-Pay-02	N
OBS-F-023	Update book stock quantity and reorder level	Administration / Inventory	TC-Inv-02	N
OBS-SR-004	Secure communication for account/order/payment traffic	Communication	TC-Sec-04	N